

THE INSTRUMENTS

BY

DOUGLAS QUIGG

RESEARCH AND THEORY

WHAT A MEASUREMENT CAN SEE, WHAT IT CANNOT, AND HOW EACH NULL WAS
DRAWN

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WHAT THE INSTRUMENTS CANNOT SEE

Purpose: Record what was measured, at what scope, and which instruments were found unable to see what they were pointed at, because a battery of null results is worth only the sensitivity of the battery. **Scope:** `maint/audit/*.py`, and the `pi` arm in `maint/audit/compressibility.py` that bounds all of it.

A night of measurement produced a stack of null results about SHA-256 and one finding that outweighs all of them: **the digits of pi pass every test in this tree, and the digits of pi compress to about a kilobyte.**

That is not a remark about pi. The battery assembled here checks for uniform bit shares, flat autocorrelation, full algebraic degree, no co-variation between output positions, and incompressibility to LZMA. A sequence with a short generating program sails through all of it. Reporting *no structure found* on the strength of it says less than it appears to.

1.1 THE INSTRUMENT FAULTS, FIRST

Five, all in the same direction, every one producing an apparent finding, every one caught by a result being too clean and never by care beforehand.

A statistic read off the top of a sort. The deepest timestamp reversals appeared to cluster near 280 seconds. They were the eight largest values of a sorted list, which are close together by construction. The full population shows no band.

Three noise floors derived instead of drawn. A shuffle of hour *labels* that returned identical counts every draw and printed a standard deviation of 0.00. A per-pool phase bar set at two Poisson floors when the statistic was a maximum minus a minimum over twenty-four bins, which spans three and a half to four and a half by construction. Every pool cleared a bar sitting below chance, and twenty hours of random trough hours read as geography. And a window-scatter floor from chi-square's variance carried through a logarithm, giving 0.0582 where shuffling gives 0.1015.

A generator with a period. Survey arms built as $at \times (2 \text{ arm} + 37) + 11 + 97 \text{ arm} \bmod 256$ are linear in the arm index, and both terms wrap together every 256 arms: arm 0 and arm 256 produced byte-identical headers. Nine hundred thirty one files held two hundred fifty six distinct arms, up to four copies each, and the duplication inflated the summed squared z from 256 to 942. That read as a thirty sigma discovery and reproduced the duplication factor to within one percent. Headers are now hashed from the arm index instead of stepped from it.

A test whose answer was geometry. Reading a state through a mirrored helix appeared to test the handedness of SHA-256, which is chiral by design: six rightward rotations and no leftward one. But reflecting a point set negates every longitude, conjugating every coefficient and negating every phase, for any data whatever. The result was fixed before the data was consulted and came back at 10^{-16} , which is float64's floor.

A scalar standing in for a structure. Per-degree power is invariant under all of $SO(3)$, and reporting it alone implies that magnitude and phase separate cleanly. They separate only on rotations about the reading axis. Under a general rotation the coefficients mix, magnitude redistributes across orders inside a degree, and the summed power is preserved because that mixing cancels and not because nothing moved. Summing over orders makes the mixing invisible.

1.2 WHAT WAS ESTABLISHED, AT ITS REAL SCOPE

On SHA-256, bounded and not proven flat. Across 550 billion evaluations spanning 256 genuinely distinct headers, the output bits are individually unbiased to about four parts in a million and carry no co-variation a permutation null cannot produce. A Walsh-Hadamard sweep over all 262,144 input masks per output bit finds no linear approximation surviving on nonces it did not see; every in-sample maximum sits below the null's own expected maximum. The algebraic degree in the nonce bits saturates by round eight, and no transform has fewer coefficients than points to ride. Reversing every rotation gives a function statistically identical to the standard one: avalanche means differ by a third of a standard error, and both reach half the output at round eighteen. Handedness is cosmetic, and the rotation amounts carry the diffusion.

Every one of those is a bound with a domain, and not an absolute. A finite observation of an unbounded input space can bound an effect and can never establish absence.

On the block header, a real and large signal. The sixteen bits BIP320 [4] reserves for version rolling give away 1.82 bits of collision entropy, 588 times the floor chance produces at that width and depth, and 45.7 percent of what the field could carry. Per-bit shares run from -5 to -51 standard errors against a swept counter's prediction, and the whole distribution is bimodal with spikes on exact powers of two: conventions, and not a sweep. The leak also moves, at $p = 0.010$ against a shuffle null, which reads as software being deployed and retired inside a forty eight day window.

On the clocks, a physical effect cleanly separated. Timestamps in 2.98 percent of blocks precede their own parent's, by up to 395 seconds. The clocks tick correctly: low-bit shares reach 3.10 against a calibrated null of 3.35, and residues mod ten give chi-square 9.9 on nine degrees of freedom. The mining process is a clean Poisson at $sd/mean = 0.9879$. The effect is therefore offset, and neither drift nor granularity, and the magnitudes are four orders past anything NTP produces.

And the separation that ties them together. The tilt lives in the fields miners *choose* and is absent from the field the function *produces*. Across 178 testable digest positions, conditioned per block on its own target, the loudest reads +2.78 standard errors against a 3.22 bar. Same eighty bytes, same blocks, same miners.

1.3 WHAT HAS NO INSTRUMENT

A one-way function is the gap between a description of about a thousand bits and a search costing 2^{48} , and nobody has proven one exists. SHA-256's hardness is conjectural. Every measurement above assumed the gap and confirmed its local consequences, which is circular, and the pi control exposed the circle.

Kolmogorov complexity is uncomputable, and no tool decides it. A compressor gives an upper bound: compression proves structure, failure to compress proves nothing, because a compressor hunts repetition and a generating program is not repetition. Measured directly, pi compresses to 1.0010 against random's 1.0001. The positive control fails completely, and any claim of the form *it does not compress, therefore no structure* is a statement about zlib.

What separates pi from noise is neither distribution nor compression. It is reproducibility: an independent method computes the same digits, and agreement between two programs sharing no arithmetic demonstrates a generating program. That is a predictive test, and the predictive test for a hash is the linear approximation search, the one axis above that pi would fail.

1.4 READING THE INTERIOR INSTEAD OF THE BOUNDARY

Every measurement above reads a digest, the boundary of the computation. That is the right shape for a physical system, where the interior cannot be reached and has to be inferred from what escapes. It is the wrong shape here, and the reason is worth stating plainly: nothing about this interior is hidden. The schedule words, the eight state words at every round, and the six values built inside a round are all readable at every instant. Those six are s_1 , ch , t_1 on one chain and s_0 , maj , t_2 on the other. The boundary framing was self-imposed.

Read that way, one flipped input bit shows something the digest cannot.

intermediate	first round it moves at all
s_1, t_1, s_0, t_2	2
ch, maj	3

The nonlinear functions lag the linear ones by a round, and the lag persists: at round seventeen the linear values sit at 15.5 to 16.0 bits of difference while Choose and Majority sit at 14.3 and 14.8.

The mechanism is in their definitions. Choose is $(e \wedge f) \oplus (\neg e \wedge g)$. Where f and g agree at a bit position the result does not depend on e there at all. Majority cannot be moved by one of three inputs either. The nonlinear functions therefore absorb the first arrival and pass it on only once a second input has been reached. A digest sums all six paths into one number and cannot say which carried what.

1.5 THE RULES, WHICH ARE A DIFFERENT KIND OF STATEMENT

Everything above is a frequency, and a frequency needs a null, a bar and an argument. The support does not. For a single-bit input difference, an output position either moves or it never does,

and a position that never moves in six hundred excitations is not rare. It is forbidden, and every trajectory requiring it is ruled out outright.

rounds	reachable of 256	forbidden	share
1	0.0	256.0	100.0%
3	7.3	248.7	97.1%
5	24.4	231.6	90.5%
7	42.0	214.0	83.6%

At round seven, 214 of 256 positions are forbidden to a single input bit. The support grows about nine positions a round after the third, putting the grammar's closing near round thirty, and that is consistent with the avalanche independently reaching 126 bits moved by round twenty. The function therefore spends roughly its first thirty rounds before every input can reach every output, and its remaining thirty-four past the point where the support constrains anything.

This is the sharpest result here and the only one needing no statistics to stand up. Its limit is equally sharp: forbidden transitions prune trajectory *searches*, as differential cryptanalysis does. Mining evaluates forward and searches no trajectories. The grammar does not help there.

1.6 TWO RESPONSE MEASUREMENTS WHOSE VERDICTS WERE WRONG

Recorded because the faults share a shape with the others and both were caught after the fact.

Susceptibility reported *clears* and did not. The loudest cell of the real 512×256 table read -4.33 standard errors against a null table's loudest of -4.11 , and the code called that a clearance. A maximum over 131,072 cells is itself a random variable with real spread. One draw of it is not a bar. The seventh derived threshold in this work, again too low.

The pair response has no power at all. Flipping two input bits and comparing against the exclusive-or of the two single differences gives 128.012 bits, and the null arm gives 127.938. Both are pinned at 128, which is half of 256 and therefore complete decorrelation. The statistic is saturated in both arms and cannot discriminate. The commentary written alongside it called the excess *the nonlinearity*, implying it carries information. It does not.

And a lead that dissolved at depth. At 1,200 samples the per-bit batch-to-batch swing had a loudest of 1.394 with the top swingers uniformly cold, which looked like a second channel invisible to a heat map. At 60,000 samples the null arm's own loudest is 1.3965, no bit clears it, and the correlation between swing and heat is -0.0671 inside a band of 0.1250. The anticorrelation was the artifact, and no signal remains.

1.7 THE TOOLS

tool	what it answers
<code>compressibility.py</code>	the pi arm, permanently; no null here escapes it
<code>walsh_bias.py</code>	every input mask by transform, and not a sampled few
<code>handedness.py</code>	SHA-256 against its mirror, rotations reversed
<code>spectral_bench.py</code>	invariance shown alongside the mixing it cancels
<code>comb_arms.py</code>	pooled depth and sign agreement, separated
<code>agglomerate.py</code>	co-variation, which every other test here is blind to
<code>chirality.py</code>	a placement validator, its actual function
<code>compare_corpus.py</code>	the clean survey against the contaminated chain
<code>collision_entropy.py</code>	every reading in one comparable unit
<code>watchdog.py</code>	calibrated models instead of guessed thresholds
<code>propagation.py</code>	one flip through every interior value, not just the digest
<code>grammar.py</code>	the support: what is forbidden, needing no null
<code>response.py</code>	susceptibility, pair response, decay; two verdicts corrected
<code>anisotropy_detector.py</code>	flat baseline, drawn bar, names the angular scale
<code>sniffer.py</code>	two stages over disjoint classes, validated before believed
<code>sniffer_bakeoff.py</code>	any new scanner proves itself on planted structure first
<code>relation_search.py</code>	integer relations, gated both directions
<code>bbp_search.py</code>	formula search, gated by rediscovering pi's

Of the first sixteen, `collision_entropy.py` is in `maint/chain/` and the other fifteen are in `maint/audit/`. The last two are in `examples/proofing/`.

1.8 THE ONE MEASUREMENT THAT CAN RETURN A DISCOVERY

Everything above returns a bound or a null. The relation search returns a statement: a closed form either holds or it does not, and anyone can check it in an afternoon without repeating the work.

Its control is the 1996 BBP formula for pi [2]. Pointed at base sixteen, power one, stride eight, the pipeline returns an exact relation, and the known formula holds on the same values to 895 bits of 900, the remainder being the series truncation. The relation is not unique: Bailey's compendium [1] notes in its Section 11 that the auxiliary sums satisfy zero relations among themselves. The exact relations form a lattice, and the known formula is one member. Demanding a particular member was the wrong control and reported a correct answer as a failure.

Two regions are closed and should not be searched. Borwein, Galway and Borwein [5] proved in 2004 that pi has no degree-one BBP formulas in a power-of-two base beyond the three already known.

FAILURE MODES, AND THE SUBTRACTION THAT REMOVES EACH ONE

Purpose: Hold the apparatus to the same subtraction the data is held to, because every fault this tree has found made a result look better than the truth and none made it look worse. **Scope:** fifteen modes, each from an instance caught here, with the detector that removes it and the detector's status.

This is not a checklist. It is a sieve, and the operation is subtractive: each detector removes a class of explanation for a number, and what is left after every subtraction is the residual. The residual is the finding. That is the same operation as the sigma-delta null, measurement minus what a null would have given, applied to the apparatus and not to the data.

Fifteen modes are known, all of them from instances actually caught in this tree. **Every one of them made a result look better than the truth.** Not one erred toward a weaker finding. That asymmetry is itself the strongest reason to run the sieve instead of trusting a number.

Status is one of **built**, where a mechanical detector exists and has been run tree-wide; **partial**, where a detector exists but has been applied only where a defect was already suspected; and **open**, where the mode is known and nothing yet catches it.

2.1 ABSENCE READ AS MEASUREMENT (BUILT)

Measuring a quantity that structurally cannot depend on the input, and reporting the resulting zero or perfect agreement as a result.

Caught. H6 rotated the nonce, which enters at W[3] in round three, instead of the chaining value, and reported a decay that never happened. H23 measured through W[17], which does not depend on W[3], and reported 100.000% agreement, which was zero equaling zero. `bench_closeness` labeled rounds 1 to 3 *premise holds: no* when the nonce had not entered yet.

The subtraction. `bench_reach.cpp`, computed and not sampled. Every state bit carries the set of input bits that can reach it, propagated through the real operations: rotation permutes positions, shift drops them, bitwise operations union bit by bit, and addition unions upward through the carry. Addition is the only widening operation, making this an over-approximation. A path may carry no influence, but **the absence of a path is exact**, the correct side to err on for a precondition.

What it settles. The nonce reaches no state bit until four rounds have run, and saturates both written words the moment it arrives:

rounds run	1	2	3	4	5
bits of state[0] reachable from W[3]	0	0	0	32	32

In the schedule, W[16] and W[17] carry **zero** bits of W[3] while W[18] carries all 32, correctly, since W[18] is the first word whose W[t-15] term is W[3]. Both of the defects that produced H6 and H23 are visible directly in those two tables, and both would have been caught before the measurement instead of after it.

The control, run in the opposite direction. An analysis claiming reachability everywhere would pass an absence test trivially. Both halves are needed. Over 24,000 random flips at rounds one to six:

pairs the analysis called unreachable	21,257
of those, a flip that changed the bit	0
pairs the analysis called reachable	2,743
of those, a flip that landed	35.07%

One violation in the first row falsifies the analysis outright. The second row has to sit away from both extremes, and does.

A structural fact that came free. Every one of the 256 state bits reads all 768 input bits from **round 21** on, the widest cone growing 256, 320, 352, 384, 416, thirty-two bits per round after the first two, which is one schedule word arriving each round. This tree has never had that number and it is exact.

2.2 HAND-DERIVED NULL (PARTIAL)

Comparing a measurement against an analytical reference derived by hand, where the derivation is wrong.

Caught. H19's expected-worst-of-3840 gave a 5.47 sigma keyhole that fell to 4.19 and then to nothing at 64× the samples. H24 modeled overshoot as geometric in nibbles and produced a 7.27× excess; the target is not on a nibble boundary and the true figure is 1.0076. H29 and H31 compared a max-statistic against a per-bin uniform and produced 775,000× where the permutation null gives 184.5×, inflated by about 4,200.

The subtraction. $\Delta_{\Sigma}[M] = M(x) - \mathbb{E}_{\pi \sim \Pi_{\Sigma}}[M(\pi(x))]$. A permutation null cannot be derived wrongly because it preserves the carrier's own counts by construction and not by arithmetic.

Status. Built in bench_deltanull.cpp, applied to one statistic. The keyhole scan's expected-worst and the corpus-size ratios in H1 are both hand-derived and neither has been checked.

2.3 ASYMMETRIC TREATMENT OF THE TWO ARMS (OPEN)

Applying one operation to one member of a pair and a different operation to the other, leaving the difference in treatment inside the measured difference.

Caught. H29 added K to one member and $\text{rotr}(K)$ to the other and produced a 25.8% survival that appeared to overturn H22. The same constant to both gives 0.000000%.

The subtraction. A swap test. Exchange the two arms and re-run. A statistic meant to be symmetric must return the same number; one deliberately asymmetric must say so and say why.

2.4 READING MORE OUTPUT THAN THE OPERATION WRITES (BUILT)

Caught. H30 measured all eight output words when a SHA-256 round writes only state[0] and state[4]. The six untouched words agreed trivially and the result read $P = 1.00000000$.

The subtraction. A write mask, in bench_reach.cpp beside the read set, since it is the same question about what the circuit does. Every output word is assumed a copy of every input word, and each of 2000 random trials removes the pairings that failed; what is left after every trial was carried instead of written.

rounds run	1	2	3	4
words written	2	4	6	8
words carried	6	4	2	0
share of an eight-word read that agrees for free	75%	50%	25%	0%

H30 read all eight words after one round. Three quarters of its comparison agreed before the round function was consulted. Reading only the written words needs no correction factor and no argument. It is the right read.

2.5 SAMPLE BELOW THE DETECTABLE EFFECT (OPEN)

Running an experiment whose expected count under the hypothesis is far below one, and reporting the absence of an effect as evidence.

Caught. H27 used $k = 8, 12, 16$ at 400 states, where $k = 16$ predicts 0.006 hits. Changed to $k = 4, 6, 8$ at 5000, with the detectable effect size printed alongside.

The subtraction. Every bench prints its expected effect size and its detection floor *before* its result. A null then reads as *below the floor* and not as *absent*.

2.6 ESTIMATOR CEILING SET BY SAMPLE SIZE (PARTIAL)

Reporting a plug-in estimate whose value is fixed by how many samples were drawn and not by the distribution.

Caught. The Renyi order one half plug-in is $2 \log_2 \sum \sqrt{p}$. Where every observed value is a singleton that sum is exactly $\sqrt{n}$, and the estimate reads $\log_2 n$ whatever the true entropy is. At 2,000,000 pairs the ceiling is 20.93 bits and the estimate reads 18.25.

The subtraction. Print the ceiling next to the estimate. Where they are close the estimate is mostly sample size and is not a measurement.

Status. Done in bench_deltanull.cpp. Every other sparse-sample estimator in the tree is unchecked, and Good-Turing coverage is the obvious place to look next.

2.7 DEGENERATE PAIRS (PARTIAL)

Constructing a pair meant to differ and failing to check that it does.

Caught. `bench_closeness` exclusive-ored at positions that were not distinct. Bits flipped back, and some pairs were identical. It printed *smallest output distance: 0 bits*, which reads as a collision, directly above prose saying nothing came close.

The subtraction. Count and report the pairs whose intended difference is zero. A nonzero count reports a construction defect and carries no result.

2.8 OPTIMIZER-DEPENDENT RESULT (BUILT)

A compiler folds, contracts, reassociates and hoists. A folded measurement is indistinguishable from a computed one by reading the output.

Caught. `bench_engines` found an entire legacy engine that was a constant function. The tree-wide audit then found three performance ratios that move with the build.

The subtraction. `maint/audit/audit_compiler.ps1` builds every bench three ways, `-00`, `-02`, and `-02 -ffp-contract=off`, and `maint/audit/audit_diff.py` compares outputs with stopwatch lines dropped.

Result. 19 of 22 judged benches are byte-identical across all three arms. The three that move are all ratios of measured throughputs: extranonce cost $11.1 \times / 28.3 \times / 23.4 \times$, expansion speedup $0.932 \times / 0.395 \times / 0.256 \times$, delivery search-over-verify $1.08 \times 10^{19} / 5.46 \times 10^{19} / 6.15 \times 10^{19}$.

What it does not close. That a folded constant was folded *correctly*. For `bench_renyi` that is settled separately by `maint/audit/verify_renyi.py`, which reimplements every printed prediction in another language against another libm. Nothing else has that check.

What chasing the last three stragglers turned up, which was worse than a folded constant. `kat_validation` and `bench_engines` produced no `-00` output because they segfault there. The trace:

```
vmovdqa %ymm0, -0x20(%rbp)    rbp = 0x5fe0b0, target 0x5fe090, which is 16 mod 32
```

`vmovdqa` is the *aligned* 256-bit move and requires 32. MinGW-w64 GCC at `-00` hand-aligns a pointer for one over-aligned `__m256i` local, the emitted `add $0x1f; shr $0x5; shl $0x5` being visible in the prologue, and then places another at a fixed offset from a frame pointer it never realigned. `-mstackrealign` changes the prologue not at all. **Nothing in this tree causes it and no source change here avoids it**; the instruction is available and `-02` runs it happily.

The resolution is that the `-00` arm drops `-mavx2`. `sha256_core.c` gates its vector arm on `__AVX2__` and the `#else` arm defers to the scalar reference. The unoptimized build is therefore the reference arm, the arm a fold audit wants, since it is the arm whose arithmetic the statistics read. `kat_validation` then runs `-00` clean: **24 run, 0 failed**.

And what that costs, which has to be said instead of buried. For the two benches exercising the vector arm, the `-00` column now tests different code from the other two columns. Agreement there is evidence about the reference arm and not about the vector arm. The vector arm is covered instead by `kat_validation` asserting it equal to the reference on the FIPS vectors, the genesis block, block 125552 and 1000 consecutive real headers. That is a strong check, and it

is a different check, and reading the audit table as though all three columns meant the same thing for those two rows would be the kind of substitution this register exists to catch.

2.9 PROSE CONTRADICTING ITS OWN TABLE (OPEN)

Caught. H23’s prose said the two *agree often* directly above a column reading 0.0000%.

The subtraction. Every prose claim names the number it rests on. The two can then be compared without re-deriving the claim.

2.10 CONFLATED QUANTITIES (OPEN)

Caught. bench_exact section 4 treated a characteristic, a product over one chosen intermediate path, as a differential, the enumeration over all intermediates. Only the two zeros in that table are informative.

The subtraction. Naming discipline. Two quantities that differ get two names, and the name records which one is meant.

2.11 PARTIAL STANDING IN FOR THE WHOLE (PARTIAL)

Caught. Reduced rounds in five benches, one output word of eight, a 64-bit prefix of a 256-bit digest, one frame, one constant, chunk two without the header, sampling in place of enumeration. Every one of these is a projection reported as the thing.

The subtraction. Lead with the whole, and label a partial as partial in the same breath. bench_closeness and bench_renyi are written that way; bench_renyi enumerates all 2^{32} nonces instead of sampling, and reads the whole 256-bit digest.

2.12 MAX-STATISTIC AGAINST A PER-BIN NULL (PARTIAL)

Reporting the largest of many bins against the expectation for one bin.

Caught. The 775,000× of H29, corrected to 184.5× by H31.

The subtraction. The permutation null, or an extreme-value reference. Drawing n samples into r bins gives an expected maximum well above n/r , and that expectation is the reference.

2.13 ARGMAX OVER A TIED SET (PARTIAL, AND FRESHLY CAUGHT)

Selecting *the most common* value where many values share the top count. The sort’s tiebreak makes the selection, and the data does not.

Caught. bench_canary steers a walk through delta space by following the most common outgoing delta. Counting how many share that maximum:

round	0	1	2	3	4	5	6
deltas tied at the peak	1	12	21	19	17	29	17

From round one onward there is no single most common delta. Between twelve and twenty-nine of them share a top count of two occurrences in 400,000 draws, and the walk follows whichever has the largest numeric value. **The comparison between the peak-steered and tail-steered walks, the entire purpose of that bench, is a comparison between two tiebreaks from round one on.**

What remains. The peak *value* is unaffected, since a tie does not change the height of the maximum, only which delta carries it. The compound figure and H3o's conclusion stand. The strategy comparison does not.

The subtraction. Count the ties at the argmax and print the count beside it. Done in `bench_canary.cpp`.

2.14 A RESULT THAT EXISTS AT ONE SEED (OPEN)

Thirty-two of the fixed seeds in this tree are the same constant. A statistic appearing at one seed and not at others is a draw, and not a finding.

The subtraction. Sweep the seed and report the spread across seeds beside the headline. The detection floor is then measured instead of derived.

Why it counts here. The three-arm compiler audit already establishes that 19 benches are deterministic: the same seed gives the same answer under three different builds. Determinism is not stability. Nothing yet tests whether the answers move when the seed does.

2.15 THE SAME FORMULA WRITTEN TWICE (BUILT, AND IT CAUGHT ITSELF)

Two copies of one statistic, differing in a case one of them handles and the other does not.

Caught, in the sieve's own instrument. `bench_renyi` had three copies of the deficit sum: the general one over bins, an inline one over the count histogram, and a third inside the Arikan block. The empty-bin rule is opposite at order one and at every other order. A bin holding nothing contributes exactly -1 at a positive order and exactly 0 at order one, since the count in front of the logarithm goes to zero faster than the logarithm goes to minus infinity. The general copy knew both rules. The histogram copy applied the first rule to order one and put **0.2965 in a table where 0.8272 belonged**, on the headline full-domain run.

Caught again, one layer out. The Poisson prediction at order one was missing a $-\log_2(\text{mean})$ term. That term is exactly zero when the mean count is one, the full-domain case. The wrong formula **agreed perfectly at the only point it was being read at** and disagreed everywhere else. A defect that hides at the operating point is the worst kind to leave in, and it was only visible because a smoke run at a smaller domain existed.

The subtraction. One function, `deficit_term`, that knows the empty-bin rule for every order, and every path goes through it. Not a check on the copies: the removal of the copies.

A limit worth stating. `maint/audit/verify_renyi.py` reimplements the predictions in another language against another libm, and it closed mode 8's remaining gap. It carried the same missing $-\log_2(\text{mean})$ term. An independent implementation written by the same author reproduces the author's misconceptions faithfully; what it catches is transcription and folding, and not a wrong idea. What caught the wrong idea was running at a domain where the term does not vanish.

2.16 WHAT THE SIEVE HAS REMOVED SO FAR

mode	detector	status	what it removed
1 absence as measurement	exact reachability, bench_reach	built	3 defects, and it predates them
4 over-reading the output	write mask, bench_reach	built	75% of H30's one-round read
8 optimizer	three-arm build diff	built	3 performance claims; 19 cleared
14 one seed	seed sweep, audit_seeds	built	H22 confirmed not a draw
15 formula written twice	one deficit_term	built	2 wrong numbers in bench_renyi
2 hand null	Π_Σ permutation	partial	a factor of 4,200 from the headline
13 tied argmax	tie count at the peak	partial	the canary's strategy comparison
6 estimator ceiling	ceiling beside estimate	partial	the order-1/2 plug-in as a measurement
11 partial for whole	lead with the whole	partial	five reduced-round claims
7 degenerate pairs	zero-difference counter	partial	a false o-bit output distance
3, 5, 9, 10, 12	none	open	nothing yet

Five modes are closed, five are partial, five have no detector. Of what remains, mode 3 (the swap test) and mode 5 (printing the detection floor before the result) are both mechanical and are next. Modes 9 and 10 are discipline instead of instrumentation and will not yield to a tool.

Mode 15 arrived last and deserves the most weight, because it was caught inside the instrument built to catch the others. The sieve does not exempt itself. Both of its instances were found by running the same code at a domain it was not designed around, the cheapest detector on this whole list, and still not a habit: **run every bench at a second scale, and compare the shape instead of the number**. A statistic that is a property of the function scales predictably; one that is a property of the apparatus does not.

AIMING THE ENGINE

Purpose: Separate the engine's capability from knowing where to point it, and record what one night of pointing it badly cost and bought. **Scope:** the arbitrary-precision engine, and the ten rules the night's six faults produced. Nothing here is about hashing.

The engine computes at a width nothing on a desktop is supposed to reach. That capability is separate from knowing where to point it, and one night of pointing it badly produced six findings that were all faults in the pointing.

3.1 WHAT THE ENGINE IS, STATED SO THE REST HAS A SUBJECT

Arbitrary-precision integer arithmetic with no floor under it, a device multiply picking its own arm by operand size, and division done by Newton so that multiplies and shifts are the only primitives anywhere. Measured on one desktop:

what	measured
pi to 10,000,000 digits	116.5 s, checked twice against an independent series
a 1,290,000,000-bit product	1.72 s of device time
2.15 trillion SHA-256 evaluations	summarized in 3 MB of counters
26.6 billion multiply-accumulates	3.2 s, once the work was in the shape the card wants

The last line rewards reading twice. The same permutation null took 24 draws a minute in Python and 2000 draws in three seconds as a Gram matrix. The card was never the constraint; the shape of the work was.

3.2 READ THE INTERIOR, NOT THE BOUNDARY

A physical system forces a boundary reading: the interior cannot be reached and is inferred from what escapes. Almost nothing the engine is pointed at has that constraint, and treating it as though it does is a blindfold put on by hand.

Pointed at a compression function, hours went into reading digests while every schedule word, every state word at every round, and every value built inside a round were readable at every instant. The interior said in one run what the boundary could not say at all: the two nonlinear functions lag the linear ones by exactly one round, because a value like $(e \wedge f) \oplus (\neg e \wedge g)$ does not

depend on e wherever f and g agree. A digest sums every path into one number and can never separate them.

Ask what the object's interior is before deciding to read its surface.

3.3 ASK WHAT IS FORBIDDEN BEFORE ASKING WHAT IS FREQUENT

A frequency needs a null, a bar, an argument and a correction. A support does not: a transition either occurred or it never did, and a transition that cannot occur rules out every trajectory requiring it, permanently, with no statistics involved.

Six hundred excitations found 214 of 256 positions unreachable at one depth. That is a harder statement than any p -value produced all night, and it took one run.

Frequencies are what is left when the rules run out. Read the rules first.

3.4 THE POSITIVE CONTROL DECIDES WHETHER A NULL MEANS ANYTHING

Without one, *we found nothing* and *we could not have found anything* print identically, and there is no way to tell them apart from inside the result.

The clearest case: a compression test found no structure in a hash and was about to be believed. Adding an arm for the digits of pi, a sequence with a *kilobyte* generating program by construction, showed pi compressing to 1.0010 against random's 1.0001. The instrument could not distinguish a known short program from none. Its verdict was empty, and the claim would have been a fact about zlib.

Every instrument gets one: pi for algorithmic structure, a planted relation for a relation search, a held-out range for anything selected, a known-answer vector for any reader, planted structure of each class for a scanner. If the control does not fire, the null is about the tool.

3.5 DRAW THE BAR, NEVER DERIVE IT

Seven derived thresholds in one night and every one came in too low. The error runs in one direction only, and that earns a rule instead of more care. Deriving a bar means enumerating the sources of variance, and the ones left out only ever *add* variance. The derived value is always the low estimate. A drawn bar has them all whether or not anyone thought of them.

the seven, by shape

a shuffle of labels returning identical counts every draw, sd 0.00
 a bar at two Poisson floors for a statistic spanning three and a half by construction
 chi-square's variance through a logarithm: 0.0582 where shuffling gives 0.1015
 $\sqrt{2 \ln N}$, the *expected* maximum, which half of all nulls exceed

Resample instead. Shuffle the labels, redraw the counts at the same N and the same bin count, recompute the identical statistic, read the percentile. Where the statistic is a maximum, a minimum or a range this is not optional, because those have no usable closed form.

3.6 MATCH THE NULL TO THE NUISANCE PARAMETERS

A null that does not match reports the nuisance parameter and calls it structure. A lit set's spectrum depends heavily on how many bits are lit, and the weight wandered between 57 and 140 across the rounds being compared. A null drawn at a fixed weight would therefore have reported that wander as signal at every step.

Hold fixed whatever the statistic is sensitive to and does not mean.

3.7 NORMALIZE UNTIL THE BASELINE IS FLAT

For points with no preferred direction every mode carries the same expected power, and a degree holds $2l + 1$ of them. Per-degree power therefore *rises* as $2l + 1$ with nothing in it at all. Drawn: the ratio is flat at 3.867 for a weight of 64 and 5.095 for 128, to within two and a half percent across ten degrees, and the two constants stand in the ratio the finite-population correction predicts, 1.318 measured against 1.333 expected.

Plotted raw, every spectrum rises and the rise is entirely mode count. Divided by $2l + 1$ the null is a horizontal line, the baseline is one number instead of a curve, and a departure is a departure with nothing to subtract by eye.

Find the form in which the null is flat, then plot that.

3.8 KNOW WHICH CLASS AN INSTRUMENT CAN SEE

Scanners were raced against planted structure of four kinds. The harmonic expansion beat every rival on anything with a shape, 328 standard errors against 14.8 for the next best, and on a forced parity it read 2.43 against its own 3.75 floor, *below* its noise. Not worse: blind, because the expansion is a linear map of the lit set and a parity is not a linear property of which bits are lit. The scan that caught it was the crudest in the table.

Best instrument is therefore not a property a single transform has. Run one cheap and complete scan over the spatial class and a second over the algebraic class, and report which fired.

3.9 DISTINGUISH THE FORMAT'S FLOOR FROM THE OBJECT'S

A residual *stopping* at 10^{-16} is float64. A residual *falling* with the width belongs to the object. Only fixed point can tell those apart, and the difference is not academic: an invariance measured in a double reported 10^{-14} and could not say whether the theorem held exactly or merely sat where a double runs out. Recomputed in fixed point it read 2^{-60} , 2^{-92} and 2^{-123} at widths of 64, 96 and 128, tracking the width. The invariance is exact.

Measure the same thing at two widths. If the answer moves, it was the format.

3.10 WIDEN EVERYTHING THAT GROWS, AND NOTHING THAT IS DEFINED

A mod 2^{32} ring is the function; widening it computes a different one and fails silently. But the statistics computed *over* that ring grow without bound and must not share its width. An accumulator held in a thirty-two bit integer was nine bits from silently wrapping, in analysis code, while the argument for keeping the ring narrow was being made about the same number.

The definition fixes a width. Everything else is arbitrary precision.

3.11 THE INSTRUMENT IS THE FIRST SUSPECT

Six faults in one night, every one producing something that looked like a finding, every one caught by the result being too clean and never by care beforehand:

what it looked like, and what it was

a thirty sigma excess that was a generator with a period of 256
 twenty hours of geography that were twenty random trough hours
 a swing of 1.394 that was the null's own largest
 a band at 280 seconds that was the top of a sorted list
 a chirality result fixed by geometry before the data was read
 an invariance that was a scalar hiding the structure it summed over

When a result is surprising, the tool produced it and the change lies in the tool. Check it first, and build the check into the tool so the next surprise has to pass it.

3.12 A MEMORYLESS CORPUS, AND WHAT THAT BUYS

A corpus is memoryless when its items carry nothing about each other: drawing one tells you nothing about the next. Establishing it is worth doing early, because it decides which whole families of method are available and which are wasted effort.

It was established here by direct measurement instead of assumption. Distance-to-target autocorrelation across 200,000 consecutive items was inside the null band at every lag from 1 to 1024. Hill-climbing on that distance lost to random draw by 3.81 standard errors on an equal budget. Not tied, *lost*, because it spent its budget probing neighbors that carried nothing.

3.12.1 *Ruled out, at and above a measured floor*

Every ordering strategy at once. If items are independent, search order cannot change the expected cost, and any cleverness about *which* item to try next is spending effort on a quantity that does not exist. One measurement retires the entire family instead of each member separately, the strongest thing a null does anywhere in this tree, and the reason it got quoted past its evidence.

Both arms read the digest through the leading-zero count, about two bits of entropy per evaluation, discarding everything below the first set bit. That was never wrong, and it was never bounded either: *we found no gradient* was standing in for *no gradient exists* with nothing fixing

the difference. Fading an injected gradient until the same statistic loses it puts a number on it: 0.0625 leading-zero bits per single-bit step, with the control flat at every rung. Above that, excluded. Below it, or expressed anywhere else in the digest, not looked at.

The engineering conclusion survives intact, because mining's objective *is* that count. The general claim did not, and the difference carries the lesson: this null was the result carrying the most weight here and the last one to get a floor, for the reason every fault in this chapter shares. It looked clean, and a readout guaranteeing a near-null looks the same as an object with nothing in it.

3.12.2 *Granted: the arms*

Independence lets observations be pooled by addition with no calibration step. A memoryless corpus can therefore be sampled in any order, at any depth, by any number of observers, and combined exactly.

This half is the useful one and it is easy to miss. Because each arm carries its own exact zero, a count c out of N deviating by exactly $2c - N$, arms of *different* depths pool correctly, the deviations add, the variances add with them, and there is no fitting anywhere. Nothing has to be planned in advance: short arms while exploring, long arms where something looks interesting, all combined into one exact estimate.

Memory per arm is flat. The counters are the same size whatever the depth, and a thousand arms therefore hold 2.15 trillion evaluations in three megabytes, about 700 million per kilobyte, and the footprint does not grow. That makes *keep looking until the question is answered* a real option instead of a wish: the record never fills.

And the two things arms buy do not scale alike, which decides how many to run:

what it buys	how it scales
pooled depth	as $\sqrt{k}$, the slow axis, traversed at whatever rate the device runs
agreement	as 2^k , each further arm halving the chance a run of agreeing signs is coincidence

The second saturates. At 32 arms a unanimous reading happens by chance about once in ten million over 256 cells, and past that point more arms buy depth only. Thirty-two is the knee.

3.13 ELISION, AND HOW TO USE IT INSTEAD OF FIGHTING IT

Elision is a value ceasing to depend on an input, the state becoming identical whether that input was one thing or another. It is normally described as loss, and from the inversion side it is. From the measurement side it is the most useful thing in the machine, because an elided dependency is a *hard* fact about the object and costs nothing to establish.

The support measurement is elision read forwards. Six hundred excitations, and 214 of 256 positions never moved at all, each one an elision, each one ruling out every trajectory that would require it, and none of them needing a null or a bar or a p -value. A frequency needs all three and most come back empty anyway.

Three ways to use it:

As a constraint, and not a loss. Where a dependency has been elided, everything downstream of that dependency is decided. That prunes searches over trajectories outright instead of probabilistically, and it makes a support measurement worth more per run than any frequency.

As a clock. Elision accumulates at a rate, and the rate is a property worth knowing. Here the support grew about nine positions a round and closed near round thirty of sixty-four, saying the object spends half its work reaching the point where everything can affect everything and the rest past it. That number is invisible to any reading of the finished output.

As the thing that tells you where to stop measuring. Once a dependency is elided no instrument recovers it, however precise, because the state is genuinely identical either way. A measurement aimed past the elision point is measuring nothing and will return a floor that looks like a result. Knowing where elision completes is knowing where the readable window closes, and in the case measured here it closed at round thirty while the tempting place to look was round sixty-four.

The general form: **elision is where the object stops being able to tell you something, and finding that boundary is cheaper and harder-edged than anything on either side of it.**

3.14 THE ORDER THESE GO IN

1. State what the object permits before measuring what it does.
2. Read the interior if the interior is readable.
3. Run the positive control. If it does not fire, stop.
4. Draw the bar. Match the null to the nuisance parameters.
5. Normalize until the baseline is flat.
6. Measure at two widths and see whether the floor moves.
7. When something clears, suspect the instrument, and only then the object.

THE DIMENSIONAL TRANSFORMS

Purpose: Record what each transform in this work translates between, what the translation buys, and the single condition under which any of them helps at all. **Scope:** the bitpack, fixed point, the number-theoretic transform, the four-step fold, the golden placement, the harmonic expansion, and the permutation read as an orthogonal map.

Seven transforms carry this work. They look unrelated and they do one thing: they express the same information in coordinates where a question is easier to ask. No transform here creates information or destroys it. Each has an inverse or is lossless at full rank.

4.1 THE SEVEN

The bitpack. An integer and a run of 32-bit limbs are the same bytes named twice. `to_bytes` and `from_bytes` are not conversions; they are two vocabularies for one object. This lets a number be handed to a device that only knows arrays, without anything being computed.

Fixed point. A real to whatever depth is paid for, held as an integer scaled by a power of two. The reals are the human coordinate and the integer is the machine's, and the translation is exact in both directions. The whole floor question turns on this: a residual stopping at 10^{-16} is speaking float64, and one falling with the width is speaking the object.

The number-theoretic transform. Coefficients to evaluations and back. A polynomial of n coefficients evaluated at n points costs n^2 pointwise and $n \log n$ through the transform, because the transform shares work between the points that pointwise evaluation repeats.

The four-step fold. One dimension of length N becomes two of $P \times Q$, with a twiddle between them. Nothing is lost, since it is a relabeling of the same indices, and it lets a transform too large for one pass fit in two.

The golden placement. An index becomes a direction. Index k goes to height $1 - 2(k + 0.5)/n$ at longitude ky , and 256 bit positions become 256 points on a sphere with no seam and no pole pile. A one-dimensional list becomes a two-dimensional surface.

The harmonic expansion. A set of points on that surface becomes $(L + 1)^2$ coefficients sorted by angular scale. At full rank it is lossless, making it a change of vocabulary and not a summary, and it buys the sorting: structure at a scale concentrates in one band instead of smearing across 256 numbers.

The permutation as an orthogonal map. k observations of a quantity become a vector in k dimensions, and correlation becomes the cosine of an angle. A permutation is then a rotation

within the coordinate subgroup, preserving every length and randomizing every angle. This one was hiding in plain sight all night as *shuffling the data*.

4.2 WHAT THEY PRESERVE AND WHAT THEY MOVE

Every one preserves the information and moves the coordinates. That family resemblance has a consequence worth stating flatly: **a transform cannot make a question answerable that was unanswerable. It can only make an answerable question cheap.**

The NTT makes the engine faster and leaves its capability where it was. The harmonic expansion makes the boundary reading legible and adds no information to it. Nothing in this tree computes anything a slower arrangement could not compute, and stating that plainly keeps the rest of the claims defensible.

4.3 THE SINGLE CONDITION: RANK DEFICIENCY

A transform pays when the object has fewer coefficients than it has points. That is the entire mechanism, in every case.

Multipoint evaluation is fast because a degree- n polynomial has n coefficients however many places it is evaluated. The work at one point overlaps the work at the next. Harmonic sorting is useful because a smooth field has most of its power in low degrees, and the high ones can be dropped. The arms pool because independent observations share a mean, and k of them carry one number's worth of uncertainty instead of k .

That fixes when a transform is useless, and the test for it is direct. A Boolean function of degree below k sums to zero over any k -dimensional affine subspace, and a cube of inputs therefore collapses if and only if the function is rank-deficient in that sense. Measured over 126 cells at every depth from eight rounds to sixty-four, nothing collapsed: every cell sat near 128 of 256, which is full degree.

An object at full rank in a basis has as many coefficients as points, and no change of basis helps. Not *the transform is slow*: there is nothing for it to share. No transform in this tree could therefore shortcut a hash, and every one of them speeds up arithmetic, because polynomials are rank-deficient by construction and a hash is built specifically not to be.

4.4 WHERE A TRANSLATION EARNED ITS KEEP, AND WHERE IT DID NOT

Earned it. The permutation-as-rotation reading explained why independent observations discriminate so sharply, and did it better than the counting argument it replaced: in k dimensions random vectors are nearly perpendicular, with typical cosine $\sqrt{2/\pi k}$, giving 0.051 at 245 dimensions and 0.025 at 1000. The discrimination is concentration of measure, and not sample size, and that framing predicts the right scaling where counting did not.

Earned it. The harmonic expansion beat every rival scanner on spatial structure by a factor of twenty-two, 328 standard errors against 14.8, because sorting by angular scale gives the coordinate in which a shape is one coefficient instead of 256.

Did not. The same expansion read 2.43 against its own 3.75 floor on a forced parity, below its own noise. A parity is not a linear property of which points are lit. The expansion is blind to it by construction, and the crudest scan in the comparison caught it. A transform translates into one language and is deaf in every other.

Did not. Reading a state through a mirrored placement looked like a chirality test and was a geometric identity: reflection negates every longitude, conjugating every coefficient and negating every phase, for any data whatever. The answer was fixed before the data was consulted. A transform's own symmetries can produce a result saying nothing about the object.

4.5 THE PRACTICAL FORM

1. Ask what coordinate the question is naturally cheap in.
2. Check the object is rank-deficient in that coordinate, because otherwise the move is free of benefit as well as free of cost.
3. Apply the transform, which changes nothing but the vocabulary.
4. Check the result is not a symmetry of the transform instead of a property of the object. Run the same reading on data with nothing in it and see whether the answer moves.
5. Translate back, and state the finding in the coordinate the question was asked in.

This work learned step four the hard way, twice.

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